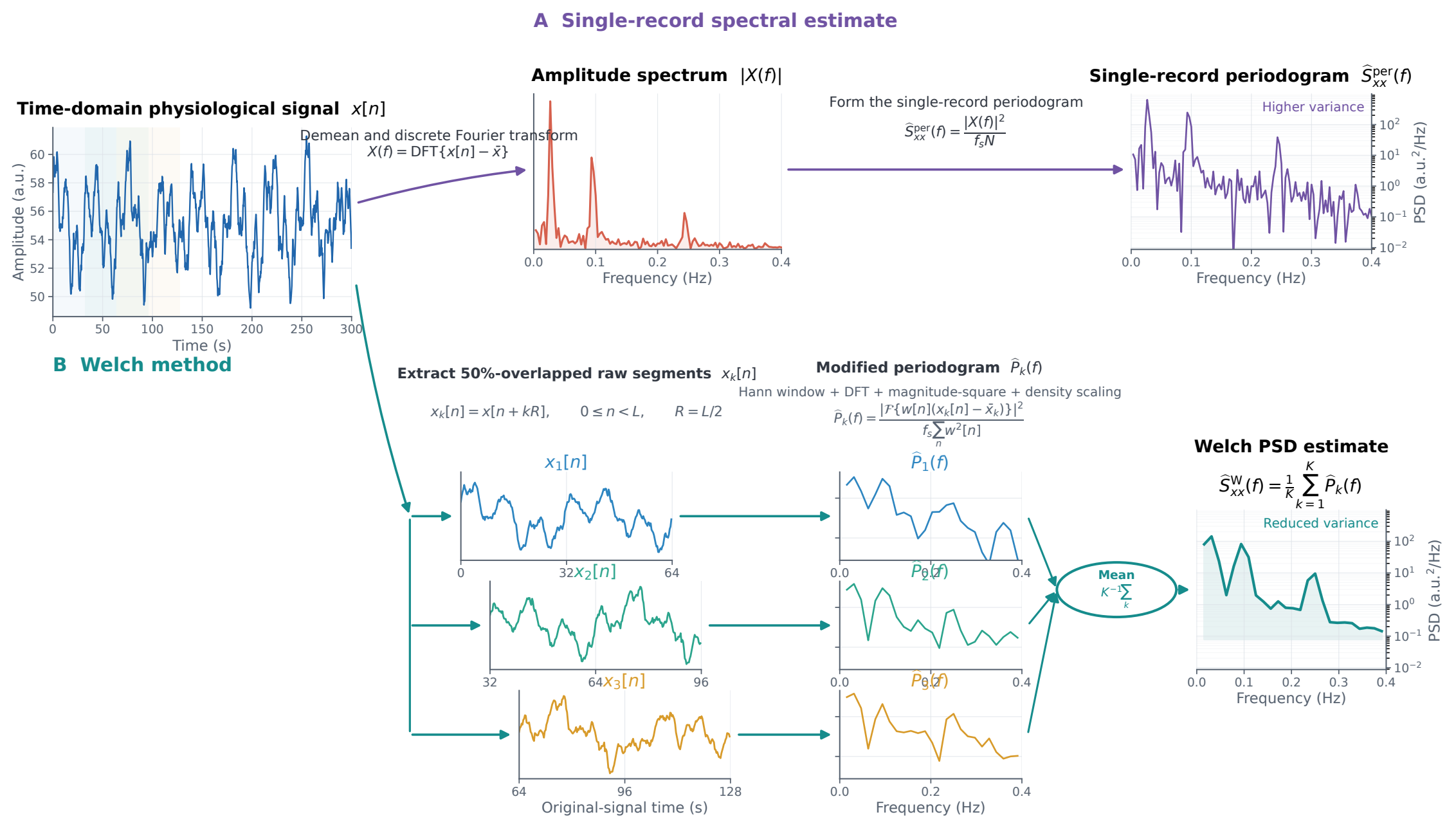

A Single-record spectral estimate
Time-domain physiological signal $x[n]$
Amplitude (a.u.)
Time (s)
Demean and discrete Fourier transform
$X(f) = \mathrm{DFT}\{x[n] - \bar{x}\}$
Amplitude spectrum $|X(f)|$
Frequency (Hz)
Form the single-record periodogram
$\widehat{S}_{xx}^{\mathrm{per}}(f) = \frac{|X(f)|^2}{f_s N}$
Single-record periodogram $\widehat{S}_{xx}^{\mathrm{per}}(f)$
Higher variance
PSD (a.u.^{2}/Hz)
Frequency (Hz)
B Welch method
Extract 50%-overlapped raw segments $x_k[n]$
$x_k[n] = x[n + kR], \quad 0 \le n < L, \quad R = L/2$
$x_1[n]$
$x_2[n]$
$x_3[n]$
Original-signal time (s)
Modified periodogram $\widehat{P}_k(f)$
Hann window + DFT + magnitude-square + density scaling
$\widehat{P}_k(f) = \frac{|\mathcal{F}\{w[n](x_k[n] - \bar{x}_k)\}|^2}{f_s \sum_n w^2[n]}$
$\widehat{P}_1(f)$
$\widehat{P}_2(f)$
$\widehat{P}_3(f)$
Frequency (Hz)
Mean
$K^{-1}\sum_k$
Welch PSD estimate
$\widehat{S}_{xx}^{\mathrm{W}}(f) = \frac{1}{K}\sum_{k=1}^{K} \widehat{P}_k(f)$
Reduced variance
PSD (a.u.^{2}/Hz)
Frequency (Hz)